

Probabilistic Sufficiency for the Obstruction Calculus: Belief-State Safety Values under Partial Observation

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Abstract

The obstruction calculus characterizes nonviability through certificates on information states; the probabilistic side asks what the certificates imply when the information state is a distribution. This paper develops the belief-state safety value $V_k(b)$ — the maximal probability of remaining safe for k steps — as a theory carried entirely in exact rational arithmetic, with six results. First, a support identity: on finite models with deterministic kernels and deterministic observation maps the posterior support coincides with the set-valued post-state of the calculus, the value-one level of the recursion coincides with the viable-set recursion $\mathcal{W}_k^{\text{fb}}$, and outside it the deficit satisfies $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$, attained on natural instances; moreover the alpha-vectors are exactly the indicators of the maximal jointly survivable subsets of the support — an antichain, Sperner-bounded — which explains why the certified witness census realizes three types. Second, exactness and parametric closed forms: over rational input data the finite-horizon values are piecewise linear with rational alpha-vectors, and on the delayed hidden-regime class both the declared hold-class and the open-loop values are proved in closed form for all initial states, all window lengths and horizons, and all prior weights — value one exactly on $z_0 \geq 1 + \min(k, T_{\text{obs}})$ and on $z_0 \geq 2$ respectively, the larger prior mass off those regions, zero below the floor — so the complete 16×12 certified value table, the $36/156$ census, and the 384 hold-direction witnesses (348 distinct after deduplication) become one-line corollaries. Third, the class lattice: declared hold $\subseteq$ open-loop $\subseteq$ sequential $\subseteq$ fully observed orders the values, and both gaps are realized exactly — the value of flexibility on the timing cells and the value of observation on the common-action cells, each $\frac{1}{2}$ at the symmetric prior and zero elsewhere. Fourth, agreement and stabilization: the segment-form enumeration reproduces the closed forms on all 48 certified cells and horizons $k \leq 4$, the certificate partition coincides with the value boundaries, the hold-class value jumps by exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$, the two classes differ exactly where $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$ (on the certified grid, exactly the timing cells, 36 combinations), and the value is frozen once $k \geq T_{\text{obs}}$ on the finite survivable-set lattice, so the certified tables are complete. Fifth, a genuinely stochastic layer: an exact weighted-path deficit identity

under noisy observation — the deficit is the prior mass weighted by fatal-reading probabilities, and a two-state sensor instance attains deficit $1/20$ against a minimal mass of $1/2$, so the min-mass bound is not universal there — a noisy-probe deadline family with three-valued exact values, and the repetition trade-off: a second probe cuts the misread probability from $1/10$ to $7/250$, and each probe step costs $1/10$ of floor margin. Sixth, a distributionally robust interpolation between the expected and adversarial operators over contamination ambiguity sets — the one-stage contaminated expectation is exact at every radius, recovering expectation at $\rho = 0$ and the worst-case reading at $\rho = 1$; a separate prior-contamination model has an exact closed form on the declared delayed hold class — an exact probe-count bound $p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2}$ giving the $\log(1/\delta)/\text{sep}^2$ probe law, and an additive deadline formula for aligned worst-loss hold-probe instances, verified on two specific parameter settings. Every closed form is verified against brute-force enumeration from the definitions, including a seeded randomized battery.

Keywords: partially observable control; viability kernels; belief states; safety values; exact arithmetic; adaptive observation; POMDP

1. Introduction

Viability under incomplete observation divides into a necessity side and a sufficiency side. The obstruction calculus treats the necessity side: certificates whose firing proves nonviability of an information state (Abaee, 2026, Obstruction certificates under incomplete observation). The probabilistic question is the quantitative counterpart. If the controller holds a distribution b over states rather than a set, how safe is safe? The answer is a function: the belief-state safety value $V_k(b)$, the maximal worst-case probability of keeping the state trajectory in the safe set for k steps. Its value-one level is the viability statement of the set-valued theory; everything below level one is measured by the deficit $1 - V_k(b)$, and the obstruction certificates reappear as lower bounds on that deficit.

This paper develops the probabilistic layer of the viability theory of Abaee (2026, Obstruction certificates under incomplete observation): the calculus there is set-valued, and the present theory quantifies what remains safe once the information state is a distribution. It brings together the stochastic selector development, the belief-

state value campaign, and the hidden-parameter learning-deadline audit into one theory with parametric closed forms, a class lattice, and an exact stochastic layer; the companion papers (the calculus, its computational certification, and the worked-systems audit) are unaffected, and nothing verified in them is re-proved differently here.

The development proceeds in seven steps. Section 2. states the augmented model — total transition rows, an absorbing unsafe state carrying its own observation, masked witnesses — and the value recursion, distinguishing the selector’s attaining set from its value-one level. Section 3. connects the belief-state recursion to the set-valued recursion of the calculus exactly, through the support of the belief. Section 4. shows the finite-horizon values are piecewise linear and exactly representable, and carries the complete certified value table over the delayed grid. Section 5. evaluates the deterministic limit, at which the recursion degenerates to a survived-mass formula over blind action sequences, with the asymmetric audit carried sequence by sequence. Section 6. reads the obstruction certificates systematically as deficit lower bounds. Section 7. proves agreement between the declared-class segment enumeration and the closed-form values on the certified delayed-regime class, quantifies what the class declaration contributes, locates the value’s jump loci, and carries the derivations. Section 8. treats observation as a design variable: probes, learning deadlines, and the exact kernel cost of late resolution of a hidden parameter. The running instances are the delayed hidden-regime instance, with the two-floor conflict as the second worked case — both shared with the obstruction calculus’s Supplementary Material. The contributions are: the belief-state value recursion with its operator laws and the selector’s viable-response structure (Section 2.); the support identity and exact sufficiency, with the piecewise-linear closed forms on the delayed hidden-regime class — hold-class and open-loop values proved for all initial states, window lengths, horizons, and prior weights — and the deficit lower-bound reading of obstruction certificates (Sections 3.–6.); and the agreement theorem between the declared-class enumeration and the closed-form values, together with the adaptive-observation layer and the distributionally robust interpolation family (Sections 7.–10.).

Relation to prior work. The belief-support reduction at the core of Theorem 3 is the finite-horizon qualitative analysis of partially observable Markov decision processes: almost-sure safety depends only on belief supports, as established for qualitative objectives by Chatterjee, Doyen, and Henzinger (2009). What is added here is quantitative and structural: the deficit bound and its witness reading (Propositions 8 and 13), the survivable-set (antichain) formula behind the witness census (Proposition 5), the class lattice with its realized gaps (Theorem 15), the parametric closed forms (Proposition 6 and Theorem 7), and the class-scoped timing obstruction. On complexity, finite-horizon partially observable problems are PSPACE-complete and their no-observation variant NP-complete (Papadimitriou and Tsitsiklis, 1987); the exponential witness-growth law of Proposition 8 is the general behaviour, and the certified two-witness families are structural rather than representative. Runtime safety filters for learning agents (Alshiekh et al., 2018) and distributionally robust partially observable models over ambiguity sets (Nakao, Jiang, and

Shen, 2021), whose value functions remain piecewise linear, bound the present development from the computational and the modelling side; Section 12. records the ambiguity-set interpolation as the formal home of worst-case wording.

All certified computations are exact: integer and rational arithmetic, standard library only, deterministic scripts (Section 11.).

2. The augmented model, the safety value, and the selector

The augmented state space. Let the physical safe set be $\mathcal{V}$ and let the state space of the model be $X^\perp = \mathcal{V} \cup \{\perp\}$, where $\perp$ is an absorbing unsafe state. For $x \in \mathcal{V}$ the transition row splits between $\mathcal{V}$ and $\perp$,

$$\sum_{x' \in \mathcal{V}} T(x' | x, a) \leq 1, \quad T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a),$$

and the bottom row is $T(\perp | \perp, a) = 1$: every row of T sums to one and $\perp$ is absorbing, so exit from $\mathcal{V}$ is absorption and no trajectory re-enters $\mathcal{V}$ from outside it. The observation model is completed on all of $X^\perp$: there is a distinguished observation $y_\perp$ with $g(y_\perp | \perp, a, \perp) = 1$ (and $g(y | \cdot, \cdot, \perp) = 0$ for $y \neq y_\perp$), while $g(\cdot | x, a, x')$ on $\mathcal{V}$ is arbitrary non-negative with the usual normalization per (x, a) . A belief is a distribution b over $X^\perp$, updated by Bayes’ rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$; because every row sums to one and every (x, a, x') emits exactly one observation with probability one, the laws $\mathbb{P}(y | b, a)$ sum to one over y at every belief — no sub-probability bookkeeping is needed. On the deterministic kernels of the certified instances this formalism converts set-valued nonviability into almost-sure nonviability on the augmented chain.

Definition 1 (safety value). Fix a declared policy class Π . The safety value at horizon k is the maximal class-survival probability

$$V_k^\Pi(b) = \max_{\pi \in \Pi} \mathbb{P}_\pi(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b),$$

the horizon counting the current state. The unrestricted class $\Pi = \Pi_{\text{seq}}$ is the class of all sequential policies.

The selector and its levels. In the selector framework this is the distributional counterpart of the recursive correspondence: the information state is b , a response is a class-admissible policy segment. Two objects must be distinguished. The *attaining set* at b ,

$$\Gamma_k^\pi(b) = \left\{ a \in A : \sum_y \mathbb{P}(y | b, a) V_{k-1}^\pi(\tau(b, a, y)) = V_k^\pi(b) \right\},$$

collects the actions attaining the value; for finite A it is nonempty at every belief and horizon — nonemptiness is automatic and carries no viability content. The *value-one level*,

$$S_k^\pi(b) = \left\{ a \in A : \sum_y \mathbb{P}(y | b, a) V_{k-1}^\pi(\tau(b, a, y)) = 1 \right\},$$

is the selector’s viable-response set, and its nonemptiness corresponds exactly to $V_k^\pi(b) = 1$. Obstruction certificates correspond to deficits $1 - V_k^\pi(b) > 0$. Notation is fixed as follows: Γ_k^π denotes alpha-vector witness sets, $\Gamma_k^\pi(b)$ the attaining action set, $\mathcal{W}_k^{\text{fb}}$ the calculus’s feedback viable-set recursion (a family of state sets), $\mathcal{W}_k$ the blind recursion (the same, but with one action for the whole set, chosen before the observation arrives), and $\mathcal{W}_k^{\text{bel}}$ the belief-space analogue of $\mathcal{W}_k$ (a family of beliefs); unsuperscripted values do not appear.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
Γ_k^Π, Γ_k^s : the alpha-vector witness sets and the attaining action set	the certificate margin $\Gamma_{\mathcal{I}}$ (certification); the recourse certificate Γ_h (obstruction calculus; min-max duals)
ρ : the contamination/interpolation radius of $\mathcal{D}_\rho$ (expected value at $\rho = 0$, worst case at $\rho = 1$)	the mesh relaxation parameter (certification); the net biomass multiplier (applied regime viability)

Theorem 2 (belief-state value recursion). $V_0^\Pi(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$,

$$V_{k+1}^\Pi(b) = \max_{a \in A(\Pi)} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k^\Pi(\tau(b, a, y)),$$

where $A(\Pi)$ is the class-admissible action set at the current stage for Markov classes Π . The recursion is exact on every finite horizon; over the unrestricted class it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

Proof. The proof is standard value iteration for the belief-state Markov decision process, with the absorbing state making V_k^Π the k -step class-survival probability (Bertsekas and Shreve, 1978). For non-Markov classes (the hold and open-loop classes of Section 7.) the class constraint couples stages; the certified computations there use the equivalent segment form of Proposition 8 below. $\square$

3. The support identity and exact sufficiency

Sperner, E. (1928). Ein Satz ueber Untermengen einer endlichen Menge. Mathematische Zeitschrift 27, 544-548.

The bridge to the set-valued theory runs through the support. Write $\text{supp}(b)$ for the states carrying positive mass, and let $\mathcal{W}_k^{\text{fb}}$ denote the calculus’s feedback viable-set recursion: $\mathcal{W}_0^{\text{fb}} = \{B \subseteq \mathcal{V}\}$, and $B \in \mathcal{W}_k^{\text{fb}}$ if some admissible a admits, for every observation y that (B, a) can produce, a $\mathcal{W}_{k-1}^{\text{fb}}$ -continuation of $\text{Post}(B, a, y)$. The blind recursion $\mathcal{W}_k$ is the special case in which the continuation may not depend on y : $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_k = \{B : \text{some admissible } a \text{ maps every } x \in B \text{ into } \mathcal{W}_{k-1}\text{-supported posteriors}\}$. Thus $\mathcal{W}_k \subseteq \mathcal{W}_k^{\text{fb}}$, strictly in general (Remark 4).

Theorem 3 (support identity). *On finite models with deterministic kernels and deterministic observation maps, for every action a and observation y with positive probability under b, a (or with unnormalized zero-mass posteriors assigned empty support),*

$$\text{supp}(b^+(\cdot | a, y)) = \text{Post}(\text{supp}(b), a, y),$$

the set-valued post-state of the calculus. Consequently, for the unrestricted sequential class, $V_k(b) = 1$ if and only if $\text{supp}(b) \in \mathcal{W}_k^{\text{fb}}$, while if $\text{supp}(b) \notin \mathcal{W}_k^{\text{fb}}$ then every policy loses a compatible state of positive mass along its realized observation path (the loss can occur at any stage), so

$$1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x).$$

Moreover $V_k(b)$ is non-increasing in k .

Proof. Support identity: a state x' carries posterior mass exactly when some $x \in \text{supp}(b)$ has $T(x' | x, a) g(y | x, a, x') > 0$; with deterministic kernels and observation maps the indicators make this set exactly $\text{Post}(\text{supp}(b), a, y)$. Sufficiency is

induction on k : the calculus’s witness sequence keeps every compatible trajectory inside $\mathcal{V}$ and the posterior supports inside the calculus’s viable sets at every stage, so every branch survives and $V_k(b) = 1$. For the converse, induct on k ; for $k = 0$ there is nothing to prove. Let $\text{supp}(b) \notin \mathcal{W}_k^{\text{fb}}$ and let π be any policy, with first action a . For each observation y that a produces with positive probability from b , the posterior $b^+(\cdot | a, y)$ has support $\text{Post}(\text{supp}(b), a, y)$ by the support identity; if that support lay in $\mathcal{W}_{k-1}^{\text{fb}}$ for every such y , then $\text{supp}(b)$ would lie in $\mathcal{W}_k^{\text{fb}}$ by the recursion’s definition. Hence for some y the posterior support lies outside $\mathcal{W}_{k-1}^{\text{fb}}$, and by the induction hypothesis the continuation of π after (a, y) loses a branch of that posterior of positive mass. Composing, π loses a branch of b of positive mass; every lost branch carries mass at least $\min_{x \in \text{supp}(b)} b(x)$, and there are finitely many branches, giving $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$. Monotonicity: absorption at $\perp$ is permanent, so a policy safe for $k + 1$ stages is safe for k (equivalently, the survivable families are prefix-nested). $\square$

Remark 4 (observation-dependent continuation is strictly stronger). Both recursions are downward closed and $\mathcal{W}_k \subseteq \mathcal{W}_k^{\text{fb}}$; the inclusion can be strict, even under the hypotheses of Theorem 3. Let $p \neq q$ be safe states and a_0 the only admissible action that keeps both alive, sending them to distinct successors $p^+ \neq q^+$ that the observation map distinguishes; let the unique safe continuation at p^+ be b and the unique safe continuation at q^+ be $c \neq b$. Then $\{p, q\} \in \mathcal{W}_2^{\text{fb}}$ — play a_0 , read the observation, continue with b or c accordingly — while $\{p, q\} \notin \mathcal{W}_2$: the first action must be a_0 , and no single second action is safe at both p^+ and q^+ . For a belief with support $\{p, q\}$ it follows that $V_2^{\Pi^{\text{seq}}}(b) = 1$ while $V_2^{\Pi^B}(b) < 1$: the value of observation is strict here in the strongest form available, all or nothing. The two recursions agree when no information arrives — on a blind window, $\mathcal{W}_k^{\text{fb}} = \mathcal{W}_k$, the set-level content of Theorem 15’s $V^{\text{ol}} = V^{\text{seq}, \text{blind}}$. Theorems 3 and 11 therefore differ in their class, not only in their normalization: the first is a feedback theorem, the second a blind one, and Proposition 6 is accordingly not a corollary of Theorem 3.

Proposition 5 (survivable sets, witnesses, and the census). *Let the kernels and observation maps be deterministic, fix a declared class, and let $\mathcal{S}_k$ be the family of subsets S of the support that are jointly survivable by one admissible class-element (sequence or policy). Then: (i) $V_k(b) = \max_{S \in \mathcal{S}_k} b(S)$; (ii) the alpha-vectors are exactly the indicators of the maximal elements of $\mathcal{S}_k$ in the componentwise order; (iii) the maximal elements form an antichain, so their number is at most $\binom{n}{\lfloor n/2 \rfloor}$ for a support of size n (Sperner); (iv) the deficit is $1 - V_k(b) = \min b(S^c)$, the minimum over the maximal survivable sets, and the min-mass bound is the special case $|S^c| = 1$. On the delayed instance the maximal survivable sets are $\{+, -\}$ exactly when $z_0 \geq 2$ and the two singletons otherwise (at $z_0 \geq 1$), giving the empty, mirror-pair, and merged witness families — the three-type census of the certified campaign is this antichain read off the grid.*

Branches survive a fixed sequence independently, survived prior mass sums over surviving branches, and maximizing over class-elements gives (i); the indicator of a survivable set is a witness and dominance gives (ii) with deduplication; maximality gives (iii) with Sperner’s theorem; and monotonicity of $b(S)$ in S gives (iv).

Proof. (i) Fix a class element; the compatible branches it keeps alive form a set $S \in \mathcal{S}_k$ by definition and its value at b is $b(S)$; conversely every $S \in \mathcal{S}_k$ is realized by some element, so maximizing over elements gives $V_k(b) = \max_S b(S)$. (ii) The element’s alpha-vector is its survival indicator $\mathbf{1}_S$; the indicator of a non-maximal S is pointwise dominated by that of a maximal $S' \supseteq S$, so the attained maximum is always attained on a

maximal element and the undominated vectors — the alpha-set — are exactly the indicators of the maximal elements. (iii) Maximality is inclusion-maximality, so the maximal sets form an antichain, and their count is at most $\binom{n}{\lfloor n/2 \rfloor}$ by Sperner’s theorem (Sperner, 1928). (iv) $1 - \max_S b(S) = \min_S b(S^c)$ over the same maximal sets; when a maximal set misses exactly one state this is the min-mass bound. The delayed-instance classification is the computed one (the verification script). $\square$

Proposition 6 (closed form on the delayed class, declared hold class). *On the delayed hidden-regime instance — states (z, θ) with $\theta \in \{-1, +1\}$, prior $b_0 = (\frac{1}{2}, \frac{1}{2})$ from z_0 , no observation before T_{obs} , exact revelation at T_{obs} , and unit drift against the floors — the belief support follows the two branches $z_0 \mp k$, and the declared hold class value (Π_{hold}) : one blind hold $u \equiv \pm 1$ for the whole window) is*

$$V_k^{\Pi_{\text{hold}}}(b_0) = \begin{cases} \frac{1}{2} (\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1]), & k \leq T_{\text{obs}}, \\ \frac{1}{2} (1 + \mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]), & k > T_{\text{obs}}. \end{cases}$$

In particular $V_k^{\Pi_{\text{hold}}}(b_0) = 1$ exactly on $z_0 \geq 1 + T_{\text{obs}}$, recovering the declared-class deterministic verdict as the value-one level; Theorem 16 quantifies where the unrestricted class does better. This is a statement about the declared class, proved by direct enumeration below — not a corollary of Theorem 3, whose value is the unrestricted one.

Theorem 7 (parametric closed forms). *On the delayed hidden-regime instance with prior mass w on the $\theta = +1$ branch (arbitrary rational $w \in [0, 1]$), initial state z_0 (arbitrary real), and effective window $\ell = \min(k, T_{\text{obs}}) \geq 1$:*

$$V_k^{\Pi_{\text{hold}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 1 + \ell, \\ \max(w, 1 - w), & 1 \leq z_0 < 1 + \ell, \end{cases}$$

$$V_k^{\Pi_{\text{ol}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 2, \\ \max(w, 1 - w), & 1 \leq z_0 < 2. \end{cases}$$

For $k = 0$, $V_0(b_0) = \mathbb{K}[z_0 \geq 1]$. At the symmetric prior both off-viability values are $\frac{1}{2}$, and Proposition 6, the complete certified table, the 36/156 census, and the witness types are one-line corollaries (the grid has $z_0 \geq 1$, so the zero level is vacuous there).

Proof. Survival requires the state in the safe set at every stage, in particular at $t = 0$: if $z_0 < 1$ both branches are below the floor immediately and the value is zero. Under a hold u both branches move monotonically, so each survives the window exactly if it is at or above the floor at its horizon position: under $u = +1$ the $+$ branch rises (safe) and the $-$ branch ends at $z_0 - \ell$, giving survived mass $w + (1 - w)\mathbb{K}[z_0 \geq 1 + \ell]$; the mirror holds for $u = -1$; maximizing over the two holds gives the first display. Under open-loop play the partial sums of any ± 1 -sequence move the branches in opposition, so the first move sends one branch to $z_0 - 1$ and joint survival requires $z_0 \geq 2$; the alternating sequence keeps both branches in $[z_0 - 1, z_0 + 1]$, which lies above the floor exactly when $z_0 \geq 2$; a single branch is survivable by the constant sequence away from it (which makes that branch rise), so off the joint region the value is the larger prior mass. $\square$

Derivation of Proposition 6. While $k \leq T_{\text{obs}}$ no observation arrives, so a declared-class policy commits to one blind hold $u \equiv \pm 1$: under $u \equiv +1$ the $\theta = +1$ branch sits at $z_0 + k$ at the horizon and the $\theta = -1$ branch at $z_0 - k$; under $u \equiv -1$ the roles swap. Each branch carries mass $\frac{1}{2}$ and survives exactly when it stays at or above the floor, giving $\frac{1}{2}(\mathbb{K}[z_0 - k \geq$

$1] + \mathbb{K}[z_0 + k \geq 1])$, and the expectation over the prior retains exactly the surviving mass — the lost branch simply carries the missing half; no adversarial selection is involved. For $k > T_{\text{obs}}$, revelation at T_{obs} splits the belief: the posterior collapses to the realized branch, and the rising control $u = +1$ keeps it safe exactly when it sits at or above the floor at revelation — so survival reduces to surviving the blind window of length T_{obs} on both branches, the mass- $\frac{1}{2}$ falling branch sitting at $z_0 - T_{\text{obs}}$ at revelation, whence the indicator $\mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]$ weighting the certainly-saved mass $\frac{1}{2}$. (The rising continuation is the point: a falling continuation would exit later, and Proposition 12 pins the frozen value for all horizons.) $\square$

4. Exact piecewise-linear values

Proposition 8 (rational alpha-vectors, class-restricted). *Fix a declared class Π . At every horizon k , V_k^Π is piecewise linear and convex: $V_k^\Pi(b) = \max_{\alpha \in \Gamma_k^\Pi} \alpha^\top b$ for a finite witness set Γ_k^Π on $\mathbb{Q}^{|\mathcal{V}|+1}$ (coordinates indexed by $\mathcal{V} \cup \{\perp\}$), constructed by the masked backup*

$$\alpha^{a, \gamma} \cdot (x) = \sum_{x'} T(x' | x, a) \sum_y g(y | x, a, x') \gamma_y(x'), \quad \gamma_y \in \Gamma_k^\Pi,$$

with $\Gamma_0^\Pi = \{\mathbf{1}_\mathcal{V}\}$ and the boundary condition $\alpha(\perp) = 0$ applied at every backup (the backup acts on $\mathcal{V}$ only; the $\perp$ -coordinate of every witness is anchored at zero and remains zero under the backup) and $\Gamma_{k+1}^\Pi = \{\alpha^{a, \gamma} : a \in A(\Pi), \gamma, \text{ a selection}\}$, duplicates removed by exact equality. For a blind open-loop window the equivalent segment form applies: each admissible action segment $(u_0, \dots, u_m) \in \Pi$ contributes the witness $\alpha^{\text{seg}} = (\mathbf{1}[\text{the segment saves branch } x])_x$ and $V_k^\Pi(b) = \max_{\text{seg}} \alpha^{\text{seg}^\top} b$. Over rational T , g , and priors, every α is an exact rational vector, every $V_k^\Pi(b)$ at a rational belief is an exact rational, and the attaining witness is part of the computation. The worst-case growth is $|\Gamma_{k+1}^\Pi| \leq |A| |\Gamma_k^\Pi|^{|\mathcal{Y}|}$; deduplication and pruned selections are the practical controls, and the exponential law is the exact complexity statement for the general class.

The PL structure is Smallwood–Sondik’s; the masking factor anchors $\alpha(\perp) = 0$ (survival requires the current state in $\mathcal{V}$, and $\perp$ is absorbing, so no backup can revive a lost branch). The exact-arithmetic discipline is the point: the backup maps rational vectors to rational vectors by finite sums and products, selections are finite, and equality tests for deduplication are exact on rationals. In the verification campaign the delayed grid enumerates the two hold-direction witnesses at each of its 192 cell-horizon combinations — 384 exact rational vectors (the two-floor instance’s witness set is carried separately) — and 348 distinct vectors remain after exact-equality deduplication.

Proof. Induction on k . At $k = 0$, $V_0^\Pi(b) = b(\mathcal{V}) = \mathbf{1}_\mathcal{V}^\top b$ with $\alpha(\perp) = 0$. For the step, substitute the induction hypothesis into the recursion of Theorem 2: the finite maximum over $a \in A(\Pi)$ and witness tuples $(\gamma_y)_y$, $\gamma_y \in \Gamma_k^\Pi$, commutes with the observation sum, and collecting the coefficient of $b(x)$ gives exactly the masked backup $\alpha^{a, \gamma}$; the $\perp$ -coordinate stays zero because the backup sums over $\mathcal{V}$ only and the anchor $\alpha(\perp) = 0$ is applied at every stage. Finiteness: finitely many action choices and witness tuples. Rationality: rational transition and observation data with rational Γ_k^Π give rational backups, by induction. Piecewise linearity and convexity are the form “finite maximum of linear functionals”; for non-Markov classes the same construction runs with history-indexed witnesses (the segment form), the class constraint pinning the available action at each stage, which restricts the backup tuples without changing the argument. $\square$

Remark 9 (dimension). On the two-floor instance the value function on the one-dimensional simplex is $\max(b_0, b_1)$ for all

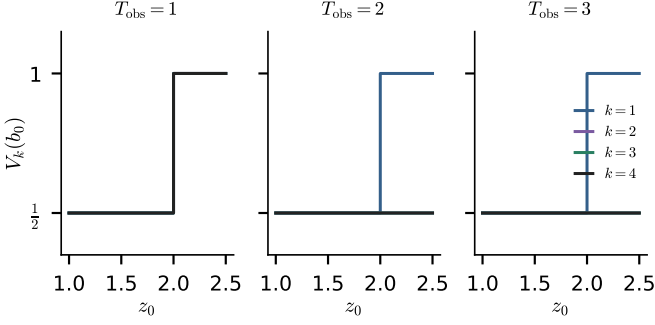


Figure 1: The closed-form values of Proposition 6 on the delayed grid: flat at $\frac{1}{2}$ while a branch falls, jumping by exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$, constant beyond.

certified horizons (Figure 5); the witness alpha-vectors live in the two-coordinate display, with the suppressed bottom coordinate identically zero — padded to $(1, 0, 0)$, $(0, 1, 0)$ on $\mathcal{V} \cup \{\perp\}$.

On the delayed grid the campaign evaluates $\alpha_k(z_0, T_{\text{obs}})$ at all $16 \times 3 \times 4 = 192$ cell-horizon combinations of $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$, $k \in \{1, 2, 3, 4\}$. The realized vectors take exactly three types: $(1, 1)$ (both branches savable in the blind window), 72 occurrences; and the mirror pair $(1, 0)$ and $(0, 1)$ (one branch savable), 156 occurrences each — the symmetry between the two mirror types is exact, $72 + 156 + 156 = 384$. The type counts are the pre-deduplication census over the two hold directions: on the 36 combinations where both holds save both branches the two witnesses coincide at $(1, 1)$ ($2 \times 36 = 72$), and exact-equality deduplication merges them, leaving $36 + 156 + 156 = 348$ distinct vectors — one per unit-value combination, two per the others. Table 1 carries the complete value table; Figure 1 displays the closed forms.

5. The deterministic limit and the two operators

Remark 10 (the two operators). The expectation over the disturbance law and the adversarial reading of its support are different Bellman operators. Proposition 11 states the deterministic degeneration as a change of operator: when the kernels are deterministic the stochastic expectation is replaced by the robust (adversarial-support) reading, and the resulting value is the robust value of the companion papers. The degeneration is a theorem about the robust operator; it is not a claim about the stochastic plant, whose expectation and robust readings differ in general (Section 12.).

Proposition 11 (survived-mass formula). *Let the kernels be deterministic, $x^+ = F(x, a, d)$, and let the disturbance be read adversarially within its support — the robust operator of Remark 10. Let the window be blind with declared sequential class Π_B . Then the degenerated value is the survived-mass formula*

$$V_k(b) = \max_{(u_0, \dots) \in \Pi_B} \sum_{x \in \text{supp } b} b(x) \mathbf{1}[x \text{ survives } (u_0, \dots)],$$

a maximum of rational linear functions of b (the surviving indicators are the degenerate alpha-vectors). Consequently: if the support B carries a jointly surviving blind sequence, $V_k(b) = 1$; and if every declared sequence loses some branch, then $V_k(b) \leq 1 - \min_{x \in B} b(x)$ — the robust common-action verdicts and the minimal-mass deficit bound of the companion papers, recovered as exact values of the robust operator.

With deterministic kernels the posterior support follows the set-valued recursion, survival from a branch under a fixed sequence is a deterministic event, and prior mass sums over surviving branches; maximizing over declared sequences gives the

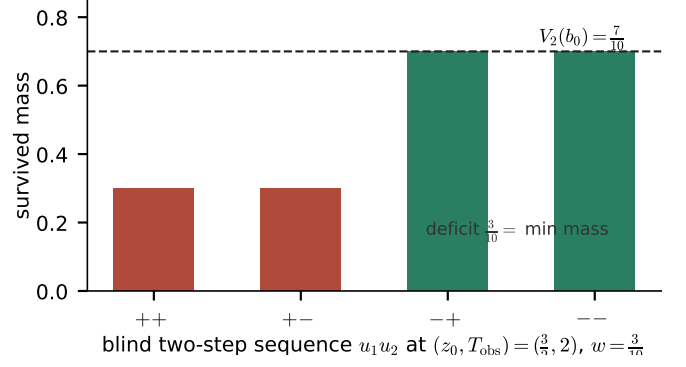


Figure 2: The asymmetric audit: survived mass by blind two-step sequence at $(\frac{3}{2}, 2)$ with $w = \frac{3}{10}$. The best sequences save the heavy branch only; the deficit $\frac{3}{10}$ is the minimal mass, attained.

formula, from which both consequences are immediate (a lost branch carries at least the minimal mass).

Proof. Under the robust operator a branch survives a declared sequence exactly when it survives under every disturbance in its support, so survival is deterministic per branch and the value is the maximum over declared sequences of the total survived mass — the displayed formula, whose summands are the survived indicators, the degenerate alpha-vectors. Each indicator is a $\{0, 1\}$ -valued linear function of b , so the value is a maximum of finitely many rational linear functions. If some declared sequence keeps every branch of B , its indicator is identically one on B and $V_k(b) = 1$. If every declared sequence loses some branch, fix a maximizing sequence and a lost branch x^* : the survived mass is $1 - b(x^*) \leq 1 - \min_{x \in B} b(x)$ — the robust common-action verdict with the minimal-mass deficit bound. $\square$

Proposition 12 (stabilization on the survivable-set lattice). *With deterministic kernels the survivable-set families satisfy $\mathcal{S}_{\ell+1} \subseteq \mathcal{S}_\ell$; hence $V_\ell(b)$ is non-increasing in the horizon and stabilizes after at most $2^{\lceil \text{supp } b \rceil}$ strict decreases, the frozen value being $\max_S b(S)$ over the maximal sets of the frozen family — an exact computable infinite-horizon value. On the delayed instance the open-loop family is frozen from $\ell = 1$ (oscillation and single-branch rises are length-independent), and the declared class freezes once $k \geq T_{\text{obs}}$ (the window ends and the rising continuation $u = +1$ then preserves safety indefinitely above the floor); consequently every certified table at $T_{\text{obs}} \leq 3$, $k \leq 4$ already records the full-horizon value.*

Survivability over a longer window is the conjunction of survivability over the shorter, giving the decreasing lattice; a monotone sequence on a finite lattice freezes; the instance's length-independence and the window's end give the two freezing statements, verified to horizon 40 by the verification script.

On the two-floor chance instance — states $\{0, 1\}$, actions $u \in \{\frac{2}{5}, \frac{3}{5}\}$, state 0 surviving exactly the lower action and state 1 exactly the higher — the one-step witness set is $\Gamma_1^\Pi = \{(1, 0), (0, 1)\}$ and $V_k(b_0) = \frac{1}{2}$ for all certified $k \leq 5$: the chance deficit $\delta = \frac{1}{2}$ is attained, and the deficit equals the minimal mass (Figure 5).

The asymmetric audit prices the deficit beyond the symmetric case. At $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ with prior masses $w = \frac{3}{10}$ and $\frac{7}{10}$, the four blind two-step sequences survive $\{\frac{3}{10}, \frac{3}{10}, \frac{7}{10}, \frac{7}{10}\}$ (Table 2): every sequence loses a branch, so $V_2(b_0) = \frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass — the degenerate bound of Proposition 13 attained off the symmetric prior (Figure 2).

Proof. A class element surviving $\ell + 1$ stages survives its prefix of ℓ stages, and declared classes are closed under prefixes,

Table 1: The complete certified value table on the delayed grid: the exact $V_k(b_0)$ (values 1 or $\frac{1}{2}$) for all 16 grid states, three blind-window lengths, four horizons — 192 cell-horizon combinations, each regenerated by script and each equal to both the declared-class segment enumeration and the closed form of Proposition 6.

z_0	$T_{\text{obs}} = 1$				$T_{\text{obs}} = 2$				$T_{\text{obs}} = 3$			
	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$
1.0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.2	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.4	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.5	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.6	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.7	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.8	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.9	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.0	1	1	1	1	1	1	1	1	1	1	1	1
2.1	1	1	1	1	1	1	1	1	1	1	1	1
2.2	1	1	1	1	1	1	1	1	1	1	1	1
2.3	1	1	1	1	1	1	1	1	1	1	1	1
2.4	1	1	1	1	1	1	1	1	1	1	1	1
2.5	1	1	1	1	1	1	1	1	1	1	1	1

Table 2: The asymmetric audit at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$, $b_0 = (\frac{3}{10}, \frac{7}{10})$: survived mass of each blind two-step sequence. Every sequence loses a branch at step 1 — at $z_0 = \frac{3}{2}$ the first move sends one branch to $\frac{1}{2} < 1$ —; the value is $\frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass.

Sequence	Survived mass	Lost branch
(+1, +1)	$\frac{3}{10}$	heavier branch, step 1
(+1, -1)	$\frac{3}{10}$	heavier branch, step 1
(-1, +1)	$\frac{7}{10}$	lighter branch, step 1
(-1, -1)	$\frac{7}{10}$	lighter branch, step 1

so $\mathcal{S}_{\ell+1} \subseteq \mathcal{S}_\ell$; by Proposition 5(i), $V_\ell(b)$ is non-increasing in ℓ . The families live in the finite lattice $2^{\text{supp } b}$, so each strict decrease retires at least one subset and at most $2^{|\text{supp } b|}$ strict decreases occur; the frozen value is the maximal-set mass of the frozen family. On the delayed instance the closed forms of Theorem 7 are length-independent in the stated regimes (oscillation and single-branch rises do not change with added stages), so the open-loop family is frozen from $\ell = 1$; the declared class freezes once $k \geq T_{\text{obs}}$ because the window has ended and the rising continuation keeps every revealed branch above the floor indefinitely; the verification script certifies the frozen values on the certified tables. $\square$

6. Certificates as deficit bounds

The obstruction certificates bound the deficit from below; their attaining witnesses are the deficit’s dual objects — computed lower-bound sides of the exact value, not surrogates for it.

Proposition 13 (deficit lower bounds). *Write $\Delta_k(b) = 1 - V_k(b)$ for the deficit, and let $\mathcal{W}_k^{\text{bel}}$ denote the k -step robust kernel in belief space: the family of beliefs whose support admits a jointly surviving declared-class sequence. Each companion certificate is a computable lower bound on the deficit. (i) Chance-constrained common action: if the one-step violation probabilities satisfy $\sum_x b(x) p(x, a) \geq \delta$ for every $a \in A$, then $\Delta_k(b) \geq \delta$ for every $k \geq 1$. (ii) Degenerate minimal mass: at the deterministic limit, if $b \notin \mathcal{W}_k^{\text{bel}}$ (the support admits no jointly surviving declared sequence), then $\Delta_k(b) \geq \min_{x \in \text{supp } b} b(x)$ — the two-floor instance attains $\delta = \frac{1}{2}$, and on the delayed grid at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ every blind sequence loses a branch, so the deficit is at least the minimal mass $\frac{3}{10}$ of the certified*

asymmetric prior, attained (Table 2). (iii) Monotone deficit: $V_{k+1}(b) \leq V_k(b)$, so the deficit is non-decreasing in the horizon; on the certified nonviability region of the delayed grid the declared-class deficit is constant in k once $k > T_{\text{obs}}$ (and at every $k \geq 1$ on the common-action cells), while on the timing cells it steps from 0 at $k = 1$ to $\frac{1}{2}$ at $k \geq 2$ (Theorem 14d). For finite horizons the exact deficit $1 - \max_{\alpha \in \Gamma_k} \alpha^\top b$ is computable, and the bounds are witnesses of it, not surrogates.

Proof. (i) At $k = 1$, survival under a has probability $1 - \sum_x b(x) p(x, a) \leq 1 - \delta$, and the hypothesis makes the bound uniform over a , so $V_1(b) \leq 1 - \delta$; monotonicity of V_k in k (Theorem 3) gives $\Delta_k(b) \geq \Delta_1(b) \geq \delta$ for every $k \geq 1$. (ii) At the deterministic limit Proposition 11 gives $V_k(b) = \max_S b(S)$ over the jointly surviving sets; if $b \notin \mathcal{W}_k^{\text{bel}}$ no full-mass set survives, so every surviving S misses a state of mass at least $\min_x b(x)$ and $\Delta_k(b) \geq \min_x b(x)$. The two-floor attainment $\delta = \frac{1}{2}$ and the delayed-instance attainments are the computed records of the verification script. $\square$

7. Agreement on the delayed-regime class

The delayed hidden-regime instance of the calculus paper — regimes $\theta \in \{-1, +1\}$, blind window T_{obs} , declared hold class $\{u \equiv \pm 1\}$, prior $(\frac{1}{2}, \frac{1}{2})$ from z_0 — is the agreement test bed. The class declaration is part of the model: the blind-window policy may be required to hold one action throughout the window (the declared hold class), or may act sequentially without observation, or may act with open-loop time-variation.

Theorem 14 (certificate–value agreement). *On the certified grid ($z_0 \in \{1.0, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$; 48 cells), over the declared hold class, and horizons $k \leq 4$: (a) the segment-form enumeration of the declared hold class (Proposition 8, segment form) returns exactly the closed form of Proposition 6 (on the certified grid the second indicator of the $k \leq T_{\text{obs}}$ line is vacuous, $z_0 + k \geq 2$; it is retained for branch symmetry). The class constraint is load-bearing here: a stagewise recursion over $A(\Pi) = \{\pm 1\}$ would compute the unrestricted value, which is 1 on the timing cells — the agreement is a statement about the segment enumeration; (b) the certificate partition coincides with the value boundaries: “viable” iff $V_k \equiv 1$; common-action-certified iff $V_1(b_0) < 1$; timing-certified iff $V_1(b_0) = 1$ but $V_{T_{\text{obs}}+1}(b_0) < 1$; (c) at the one-step horizon both mirror witnesses (1, 0) and (0, 1) attain the value $\frac{1}{2}$ below $z_0 = 2$, and the merged (1, 1) attains 1 at and above it; at horizon $k \leq T_{\text{obs}}$*

the family changes at $z_0 = 1 + k$ — an off-grid consequence of the closed form (in particular $z_0 = 3$, not 2, for $k = 2$ with $T_{\text{obs}} \geq 2$), verified by the verification script beyond the certified grid, which stops at $z_0 = 2.5$; and read as a function of the continuous parameter z_0 the hold-class value has jump discontinuities exactly at $z_0 = 1 + \min(k, T_{\text{obs}})$ — a jump of exactly $\frac{1}{2}$, from $\frac{1}{2}$ to 1 — these loci being the certificate edges on the grid (kinks in the strict sense belong to the piecewise-linear value as a function of the belief, not of z_0); (d) on the nonviable cells the hold-class deficit equals $\frac{1}{2}$ at every horizon $k > T_{\text{obs}}$ and at every $k \leq T_{\text{obs}}$ with $k > z_0 - 1$, while it is 0 at horizons $k \leq z_0 - 1$ (on the certified grid: on the timing cells the deficit is 0 at $k = 1$ and $\frac{1}{2}$ at $k \geq 2$; on the common-action cells it is $\frac{1}{2}$ at every $k \geq 1$) — the minimal-mass bound attained with the minimal prior from the first horizon at which the falling branch exits.

Derivation of the jump loci. Fix T_{obs} and k . On the closed form: if $k \leq T_{\text{obs}}$ the value is $\frac{1}{2}(\mathbb{K}[z_0 \geq 1 + k] + 1)$, the second indicator being identically one on the grid; it jumps by exactly $\frac{1}{2}$ at $z_0 = 1 + k$. If $k > T_{\text{obs}}$ the value is $\frac{1}{2}(1 + \mathbb{K}[z_0 \geq 1 + T_{\text{obs}}])$, jumping by exactly $\frac{1}{2}$ at $z_0 = 1 + T_{\text{obs}}$. Both loci are $z_0 = 1 + \min(k, T_{\text{obs}})$; between jumps both expressions are constant in z_0 . The witness family changes exactly at the same loci: the attaining one-step vectors are both mirrors $(1, 0)$ and $(0, 1)$ below $z_0 = 2$ and the merged $(1, 1)$ at and above, and at horizon k the change sits at $z_0 = 1 + \min(k, T_{\text{obs}})$. $\square$

Theorem 15 (class lattice and realized gaps). *For the value of Definition 1, $\Pi \subseteq \Pi'$ implies $V_k^\Pi(b) \leq V_k^{\Pi'}(b)$. On the delayed instance with $\ell \geq 1$ and $z_0 \geq 1$:*

$$V^{\Pi_{\text{hold}}} \leq V^{\Pi_{\text{ol}}} = V^{\Pi_{\text{seq, blind}}} \leq V^{\Pi_{\text{obs}}},$$

the middle equality holding because no observation arrives inside the window, so sequential blind policies reduce to open-loop sequences; and $V^{\Pi_{\text{obs}}} = \mathbb{K}[z_0 \geq 1]$ (the controller knows θ at $t = 0$ and plays the rising control). The gaps are exact: $V^{\Pi_{\text{ol}}} - V^{\Pi_{\text{hold}}} = 1 - \max(w, 1 - w)$ exactly on $\{2 \leq z_0 < 1 + \ell\}$ — the value of flexibility — and $V^{\Pi_{\text{obs}}} - V^{\Pi_{\text{ol}}} = 1 - \max(w, 1 - w)$ exactly on $\{1 \leq z_0 < 2\}$ — the value of observation; at the symmetric prior each gap is $\frac{1}{2}$ on its cells and 0 elsewhere, and the two cell families are disjoint.

Inclusion monotonicity is immediate from the maximizing definition; the instance values are Theorem 7's and the observed value is the rising computation; the gap arithmetic is $1 - \max(w, 1 - w) = \min(w, 1 - w)$.

Proof. Monotonicity: $\Pi \subseteq \Pi'$ makes every admissible class element of Π admissible for Π' , and the value is a maximum over class elements, so $V_k^\Pi \leq V_k^{\Pi'}$. Middle equality: inside the blind window no observation arrives, so a sequential blind policy's stage actions depend only on the observation-free history — the initial belief — and the policy is one fixed sequence: sequential blind collapses to open-loop. Observed value: knowing θ at $t = 0$, the controller plays the rising control on every branch (drift $+\frac{3}{10}$), viable exactly from $z_0 \geq 1$, giving $V^{\Pi_{\text{obs}}} = \mathbb{K}[z_0 \geq 1]$. The gaps: hold and open-loop values are the closed forms of Theorem 7; subtracting the two case displays, the gap $V^{\Pi_{\text{ol}}} - V^{\Pi_{\text{hold}}} = 1 - \max(w, 1 - w)$ survives exactly on the cases where the displays differ — $\{2 \leq z_0 < 1 + \ell\}$ — and vanishes off them; the observation gap's locus is the same case analysis against $\mathbb{K}[z_0 \geq 1]$. The script verifies both loci cell by cell on the certified grid. $\square$

Theorem 16 (class declaration). *The declared hold class and the unrestricted sequential blind class coincide at the one-step horizon. For horizons $k \geq 2$ they differ exactly where $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$ — on the certified grid ($z_0 \leq 2.5$) exactly the timing cells, since $1 + \min(k, T_{\text{obs}}) \geq 3$ there — and there*

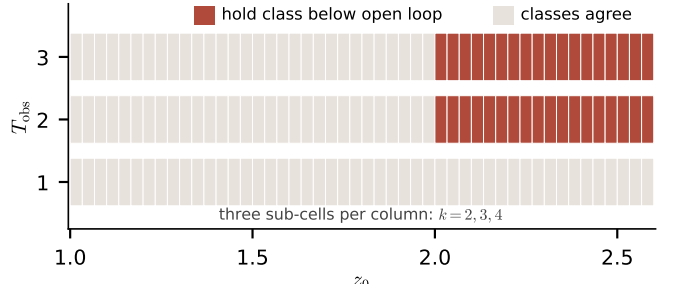


Figure 3: The class declaration on the certified grid: red cells mark cell-horizon combinations where the declared hold class and the unrestricted open-loop class differ — exactly the timing cells ($2 \leq z_0 < 1 + T_{\text{obs}}$) at horizons $k = 2, 3, 4$ (three sub-cells per column); 36 combinations in all. The classes agree at $k = 1$, on the common-action cells, and on the viable cells.

open-loop time-variation within the blind window lifts the value to one: an alternating hold sequence keeps both branches at or above the floor whenever $z_0 \geq 2$, which no constant hold does (off the grid the hold class already attains one at $z_0 \geq 1 + k$: e.g. at $(z_0, T_{\text{obs}}, k) = (3.5, 3, 2)$ both classes attain one, so no difference). On the common-action cells $z_0 < 2$ the restriction costs nothing — no blind sequence saves both branches — and on the viable cells both classes attain one. On the certified grid the difference comprises exactly the 12 timing cells at each of $k = 2, 3, 4$: 36 cell-horizon combinations in all (Figure 3).

Derivation of Theorem 16. At $k = 1$ both classes choose one blind action, so they coincide. Let $k \geq 2$ and write the two branch positions after a blind sequence $u_{1:\ell}$, $\ell = \min(k, T_{\text{obs}})$, as $z_0 \pm \sum_t u_t$. If $z_0 < 2$, some branch sits below $2 - (z_0 - 1)$ steps from its floor, and since the partial sums of any ± 1 -sequence move one branch down whenever they move the other up, one branch reaches its floor: no blind sequence saves both, both classes give $\frac{1}{2}$, no difference. If $z_0 \geq 1 + \min(k, T_{\text{obs}})$ the constant hold already survives the window on both branches (the falling branch exits only after $k > z_0 - 1$ steps) and both classes attain one. On the difference cells $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$: a constant hold drives the falling branch to $z_0 - \ell < 1$ (deficit), while the alternating sequence has partial sums in $\{0, \pm 1\}$, so both branches stay at or above $z_0 - 1 \geq 1$ through the window, and revelation at T_{obs} then regenerates both: the unrestricted class attains one while the declared class does not. This locates the difference setwise. $\square$

Remark 17 (the timing obstruction is class-scoped). The calculus paper's timing certificates — $\sigma^*(B_0) = z_0 - 1$, with the certificate firing exactly when $T_{\text{obs}} > \sigma^*$ (equivalently $z_0 < 1 + T_{\text{obs}}$; viability at the deadline holds when $T_{\text{obs}} \leq \sigma^*$) — are statements about the declared hold-until-review class — the institutional reading of a sampled-review regulator. Theorem 16 quantifies the contrast: within the declared class the timing cells are nonviable with deficit $\frac{1}{2}$ at every horizon $k \geq 2$ (and 0 at $k = 1$); under the unrestricted sequential blind class the same cells attain value 1 by oscillation, and on the certified grid no other cell differs. The timing obstruction therefore certifies exactly what it claims — nonviability *within the declared class* — and no statement of the companion papers is altered by this reading. The word “adaptive” is avoided for blind windows: while the window is blind, no observation updates the belief, and the only distinction available is hold-constant versus time-varying open-loop control.

Table 3: The pinned-probe learning-deadline audit: the exact threshold $\frac{21}{10} + \frac{T_{\text{learn}}}{10}$ and the resulting kernel on the grid at each deadline.

T_{learn}	Threshold	$ K $	Kernel K on the grid
0	21/10	5	{2.1, 2.2, 2.3, 2.4, 2.5}
1	22/10	4	{2.2, 2.3, 2.4, 2.5}
2	23/10	3	{2.3, 2.4, 2.5}
3	24/10	2	{2.4, 2.5}
4	25/10	1	{2.5}

8. Adaptive observation and hidden parameters

Observation is a design variable. Structural uncertainty — a hidden parameter resolved too late, rather than noisy data — is the obstruction: the state augments with a hidden parameter $\theta \in \Theta$, the belief lives on the product (x, θ) , and the obstruction is the emptiness of the joint floor-servable set over (x, θ) cells. On the declining instance the parameter $\theta \in \{-1, +1\}$ gates the correctable component of a $-\frac{1}{10}$ drift; the list of affected sustainability parameters — regeneration rates, climate sensitivity, institutional effectiveness — motivates the model class, and the paper verifies on exact instances.

Proposition 18 (probes and learning deadlines). *On the certified grid, with quantized control $u \in \{-1, 0, 1\}$ and hold-then-probe-then-act schedules: with θ observed, every cell is viable (16 of 16). Under one-probe endogenous learning with free probe timing (probe at the start), the probe reveals θ exactly — the branch separation is 2, so one observation splits the belief with no statistics — and survival forces $z_0 \geq 21/10$, independent of any deadline. Under the certified pinned-probe family — hold $u = 0$ to the probe horizon and probe at T_{learn} , so no earlier probe is available within the family — viability holds exactly when*

$$z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}, \quad T_{\text{learn}} = 0, 1, 2, 3, 4,$$

the pinned-probe learning-deadline identity: the deadline prices the forced late probe, and the kernel shrinks exactly linearly and antitone in it, $|K| = 5, 4, 3, 2, 1$ on the grid for $T_{\text{learn}} = 0, 1, 2, 3, 4$ (Table 3 and Figure 4). Two exact benchmarks bound the family from both sides: with free probe timing the threshold is flat at 21/10, and with probe-free exogenous revelation at T_{learn} (hold, then the learned act; no probe) viability holds exactly when $z_0 \geq 1 + T_{\text{learn}}/10$ — kernel sizes 16, 15, 14, 13, 12 on the grid for $T_{\text{learn}} = 0, 1, 2, 3, 4$; the certified family sits between the two. A two-patch instance with unknown regeneration rate $g \in \{1, 2\}$ shows the same phenomenon on the certified two-patch belief pair: unknown-regeneration nonviable, (x, g) -observed viable.

Proof. The schedule pins the probe to T_{learn} : within the family no earlier probe is admissible, so the free-probe and probe-free thresholds are not attainable in it. Before T_{learn} the drift is $-\frac{1}{10}$ (hold $u = 0$), so survival to the probe forces $z_0 - \frac{T_{\text{learn}}}{10} \geq$ the probe threshold; the probe $u = +1$ moves z by $-\frac{1}{10} + \theta$, which must stay safe on both branches, giving the threshold $\frac{21}{10}$ at $T_{\text{learn}} = 0$; after the probe the learned law $u = \theta$ has drift $-\frac{1}{10} + \theta^2 = \frac{9}{10} > 0$, so survival is indefinite. The branch positions $z_0 - \frac{1}{10}T_{\text{learn}} - \frac{1}{10} \pm 1$ differ by 2, so the post-probe reading determines θ uniquely. Antitonicity is the threshold identity; strictness on the grid is the computed counts 5, 4, 3, 2, 1. $\square$

9. Genuinely stochastic instances

Everything to this point has deterministic kernels and observations; the values are prior masses of surviving branches. This

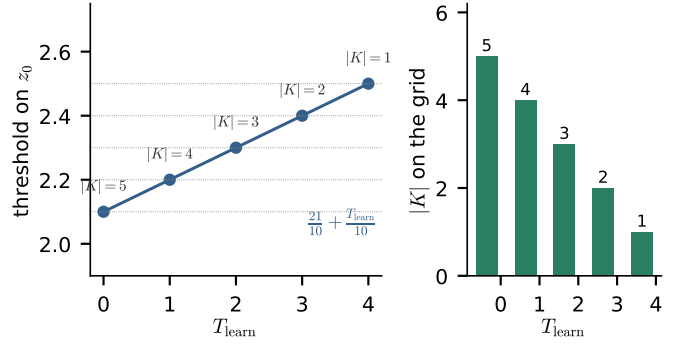


Figure 4: The pinned-probe learning-deadline identity: the viability threshold increases exactly linearly in T_{learn} (left), and the kernel on the grid shrinks linearly and antitone, $|K| = 5, 4, 3, 2, 1$ (right); the probe-free benchmark $1 + T_{\text{learn}}/10$ (kernels 16 down to 12) and the flat free-probe threshold 21/10 bound it from below and above.

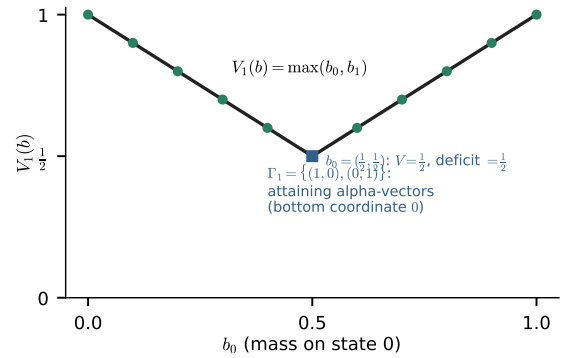


Figure 5: The two-floor chance instance: the one-step value is $\max(b_0, b_1)$, attained by the alpha-vectors of $\Gamma_1^\Pi = \{(1, 0), (0, 1)\}$ with zero bottom coordinate (green probes); at the symmetric prior the value is $\frac{1}{2}$ and the deficit equals the minimal mass.

section opens the stochastic layer, where values cease to be two-valued and the min-mass bound ceases to be universal.

Proposition 19 (weighted-path deficit identity). *At the one-step horizon, with finite X , deterministic per-reading actions $a(y')$, and arbitrary normalized observations, the deficit is exactly*

$$\Delta_1(b) = \sum_x b(x) p_x, \quad p_x = \mathbb{P}(a(y') \text{ unsafe at } x \mid x),$$

the prior mass weighted by fatal-reading probabilities. The min-mass bound is the case $p_x \equiv 1$ (deterministic readings, no common action) and is not universal: in the two-state instance with a_i safe only at x_i and sensor error ε ($\mathbb{P}(\text{reads } j \neq i \mid x_i) = \varepsilon$), the reading-own policy loses ε at every prior, while for $0 < \varepsilon < 1/2$ the optimized deficit is $\min\{\varepsilon, b(x_1), b(x_2)\}$; at $b = (\frac{1}{2}, \frac{1}{2})$, $\varepsilon = \frac{1}{20}$, the deficit $\frac{1}{20}$ is strictly below the minimal mass $\frac{1}{2}$. This delimits the deterministic-limit reach of the calculus's probabilistic development (its Section 10): there the deficit from unity is bounded below by the smallest compatible mass under adversarial disturbance readings, an estimate exact at the degeneration this paper's Proposition 11 records; the sensor instance shows the estimate is scoped to that limit, not a law of the noisy layer. Conversely, the calculus's delayed-hidden-regime instance is the deadline law of Proposition 24 at $d_0 = 1$, $e_p = 0$: its viability boundary $z_0 \geq 1 + T_{\text{obs}}$ is the law's threshold form, and the tabulated agreement there is cell-by-cell.

Proof. Fix a deterministic per-reading action map $a(\cdot)$. One-step survival is $\sum_x b(x) \mathbb{P}(a(y') \text{ safe} \mid x) = 1 - \sum_x b(x) p_x^{a(\cdot)}$, and optimizing over maps gives $V_1(b) = 1 - \min_{a(\cdot)} \sum_x b(x) p_x^{a(\cdot)}$ — the displayed identity at the optimizing map. In the two-state instance the reading-own-action map makes $p_x = \varepsilon$ for both states (the wrong reading arrives with probability ε and its action is fatal), but a constant reading-to-action map may do better at a skew prior. Enumerating the four deterministic maps for $0 < \varepsilon < 1/2$ gives $\Delta_1(b) = \min\{\varepsilon, b(x_1), b(x_2)\}$. The reading-own map is optimal at the symmetric prior, and at $b = (\frac{1}{2}, \frac{1}{2})$, $\varepsilon = \frac{1}{20}$, the deficit $\frac{1}{20}$ is strictly below the minimal mass $\frac{1}{2}$. $\square$

Proposition 20 (noisy probes and the repetition trade-off). *On the declining instance (blind drift $-\frac{1}{10}$), pin a single probe to horizon t_p whose reading is correct with probability $1 - \varepsilon$ and flipped with probability ε , then act on the reading to horizon $k \geq t_p$. With $z_p = z_0 - t_p/10$:*

$$V_k(b_0) = (1 - \varepsilon) \mathbb{K}[z_p \geq 1] + \varepsilon \mathbb{K}[z_p \geq 1 + \frac{11}{10}(k - t_p)],$$

exactly (the correct reading acts under drift $\frac{9}{10}$, the flipped reading under drift $-\frac{11}{10}$). The values realize exactly three levels $\{0, 1 - \varepsilon, 1\}$ — genuinely non-two-valued — and the min-mass bound survives here as a bound (the deficit ε or 1 dominates the post-probe minimal mass ε) while failing as an identity. With three pinned probes and majority reading, the wrong-majority probability is $3\varepsilon^2(1 - \varepsilon) + \varepsilon^3 = \varepsilon^2(3 - 2\varepsilon) - \frac{7}{250}$ at $\varepsilon = \frac{1}{10}$, against $\frac{1}{10}$ for one probe — and the same display holds with $p = \frac{7}{250}$ in place of ε : each additional probe step costs $\frac{1}{10}$ of floor margin to the blind drift, the exact explore-exploit exchange at this scale.

Proof. Condition on $(\theta, \text{reading})$: the four joint cases carry probabilities $\frac{1}{2}(1 - \varepsilon)$ and $\frac{1}{2}\varepsilon$ per branch; the correct reading acts under drift $-\frac{1}{10} + \theta^2 = \frac{9}{10}$ and survives from $z_p \geq 1$; the flipped reading acts under drift $-\frac{1}{10} - 1 = -\frac{11}{10}$ and survives exactly when $z_p - \frac{11}{10}(k - t_p) \geq 1$; summing gives the display. For the majority family, the wrong-majority event is two or three flipped readings, of probability $3\varepsilon^2(1 - \varepsilon) + \varepsilon^3$; the falling continuation is as above. The identity's proof is the law of total probability over readings at the one-step horizon. $\square$

10. Distributionally robust interpolation

Remark 10 supplies the expected and adversarial endpoints; one-stage contamination has an exact inner expectation formula, not in general an affine interpolation of multi-stage values. Fix the contamination ambiguity set $\mathcal{D}_\rho = \{q = (1 - \rho)p + \rho r : r \in \Delta(Y)\}$ at radius $\rho \in [0, 1]$.

Proposition 21 (one-stage contamination and scope). *Let $\mathcal{D}_\rho$ be as above and define $V_{k+1}^\rho(b) = \max_a \inf_{q \in \mathcal{D}_\rho} \sum_y q(y) V_k^\rho(\tau(b, a, y))$. Then: (i) the inner infimum is exact,*

$$\inf_{q \in \mathcal{D}_\rho} \sum_y q(y) v_y = (1 - \rho) \sum_y p(y) v_y + \rho \min_y v_y,$$

the adversary spending the whole contamination ρ on the worst outcome; (ii) the identity in (i) need not interpolate multi-stage values: under repeated contamination, continuation values themselves depend on ρ ; (iii) no general concavity claim follows. At $\rho = 0$ the nominal finite-action belief value is convex and piecewise linear in b as in Proposition 8.

Proof. The infimum of a linear functional over the contamination set is attained at a vertex $r = \delta_{y^*}$, giving (i). Neither a Bellman maximization nor repeated contaminated backups distributes over the endpoint mixture: with one safe state, transition survival $1/2$, a revealing safe/exit observation, one action

and $k = 2$, stagewise $\rho = 1/2$ gives survival $1/16$ rather than the endpoint interpolation $1/8$. The nominal two-floor value $V_1^0(b) = \max(b_1, b_2)$ is convex, not concave: its midpoint is $1/2$ and both endpoint values are 1. In the *different*, prior-contamination setup on the delayed class with prior contamination $q(+1) = (1 - \rho)w + \rho r$, the hold-class value is exactly

$$V_k^{\rho, \Pi^{\text{hold}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 1 + \ell, \\ (1 - \rho) \max(w, 1 - w), & 1 \leq z_0 < 1 + \ell, \end{cases}$$

the adversary pouring the contamination onto a falling branch; at the symmetric prior the three ρ -levels are 0, $\frac{1-\rho}{2}$, and 1 (verified by exact enumeration on the swept grid). $\square$

The one-stage operator and the specific prior-contamination example distinguish expected from adversarial readings; no untested claim is made that every table is an affine multistage ρ -mixture.

Proposition 22 (probe-count bound). *Let $n = 2m + 1$ pinned probes with per-probe error $\varepsilon < 1/2$ be read by majority. The wrong-majority probability satisfies*

$$p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2} \leq e^{-2n \text{sep}^2}, \quad \text{sep} = \frac{1}{2} - \varepsilon,$$

so $n \geq \ln(1/\delta)/(2\text{sep}^2)$ probes achieve wrong probability $\leq \delta$: the probe count grows like $\log(1/\delta)$ over the squared separation, and on the declining instance each probe step costs $\frac{1}{10}$ of floor margin to the blind drift (the exploration-safety exchange of Proposition 20, now with its rate). At $\varepsilon = \frac{1}{10}$ the bound reads $p_{\text{wrong}}(n) \leq (3/5)^n$, verified exactly at $n = 1, 3, 5, 7, 9$ (the exact values $\frac{1}{10}, \frac{7}{250}, \frac{107}{12500}, \frac{341}{125000}, \frac{2273}{2500000}$).

Proof. The bound is the standard misselection probability of the binomial head: $p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2}$ by pairing each wrong-majority string with its complement and Stirling's estimate, and $4\varepsilon(1 - \varepsilon) = 1 - 4\text{sep}^2 \leq e^{-4\text{sep}^2}$. $\square$

Remark 23 (noiseless limit of the probe-count law). The exact split of Proposition 18 is the $\varepsilon \rightarrow 0$ degeneration of the probe-count bound: the separation $\text{sep} = \frac{1}{2} - \varepsilon$ rises to $\frac{1}{2}$ and $p_{\text{wrong}}(1) = \varepsilon$ falls to 0, so one probe already achieves $\delta = 0$ — the branch split of separation 2 needs no statistics precisely because the bound's exponent vanishes with ε . For $\varepsilon > 0$ the count law prices what the exact split gets for free, and on the declining instance each probe step additionally pays its $\frac{1}{10}$ of floor margin (Proposition 20).

Proposition 24 (additive deadline with aligned worst losses). *Let the hidden parameter enter the drift additively: $\dot{z} = d(u, \theta) = c(u) + h(u)\theta$ with $\theta \in \Theta$ finite, floor $z \geq 1$, and consider the hold-probe-learn schedule family: hold u_0 to the probe horizon T (the probe pinned there, no earlier probe in the family), probe with u_p , then play $u^*(\theta)$ with $\min_\theta d(u^*(\theta), \theta) = \mu > 0$. Set $d_0 = \max_\theta (-d(u_0, \theta))$ and $e_p = \max_\theta (-d(u_p, \theta))$. Assume the hold and probe each take the stated durations ($T \geq 0$ and one unit), $d_0, e_p \geq 0$, and a θ^* attains both loss maxima. Then indefinite viability holds exactly when*

$$z_0 \geq 1 + T d_0 + e_p,$$

the floor plus the aligned worst losses. Without a common maximizer, the exact threshold for this fixed schedule is instead $1 + \max_\theta \max\{0, -Td(u_0, \theta), -Td(u_0, \theta) - d(u_p, \theta)\}$; positive post-learning drift preserves safety thereafter. The certified instance has $d_0 = \frac{1}{10}$, $e_p = \frac{11}{10}$, giving $\frac{21}{10} + \frac{7}{10}$; a second instance with drift $-\frac{3}{20}$ and excursion $\frac{21}{20}$ gives $\frac{41}{20} + \frac{37}{20}$ — both verified exactly at all swept (z_0, T) .

Proof. Necessity under the stated aligned-maximizer condition: the same θ^* realizes both the hold loss d_0 and probe excursion e_p , dropping the state to $z_0 - Td_0 - e_p$, below the floor exactly when the display fails. Sufficiency follows from the nonnegative worst-loss bounds on both prefix endpoints (and the linear stock path within each phase); the learned drift $\mu > 0$ then lifts it away from the floor forever. $\square$

Reciprocal check (dual-certificate companion). On the two-window residue instance of the minimax companion (Abaee, 2026, Minimax dual certificates), the interpolation family above is exact at its coupling extremum: $\sup_{q \in \mathcal{D}_\rho} \mathbb{E}_q[G] = \rho$ at every probed $\rho \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, the product coupling attaining the $\rho = 1$ end, with the one-step marginals preserved at every ρ and the coupling-selection difficulty left orthogonal to the mixture parameter; the joint check is archived with the companion’s verification record.

11. Verification methods

Every value, bound, identity, threshold, figure, and tabulated entry in this paper is regenerated by deterministic scripts in exact integer and rational arithmetic, standard library only. Three per-system scripts cover the worked instances; one master entry point executes all of them. The computational companion papers (exact belief computation and its certification) carry the scaling record — exact point-based evaluation at probed beliefs (in the POMDP literature’s sense, Lovejoy, 1991 — here exact at the probed beliefs, not bounds), the antichain compression of survivable-set lattices, and the four-parameter cube’s Hamming classification — built on the alpha-vector representation of Proposition 8; its scripts chain this paper’s seeds but are not part of this paper’s chain. The belief-state campaign script runs sixteen check families on the stochastic values (the two-floor instance, piecewise-linearity, the deterministic limit, the 48-cell agreement, the partition coincidence, the witness family, the deficit identity, the unrestricted class, rationality of all 384 campaign alpha-vectors, deficit monotonicity, the oscillation identity, model completeness, the selector split, dimension consistency, the jump loci, and the declarations), chaining two per-instance scripts of twenty-one further checks. The stochastic selector script re-derives every selector claim in fifteen check families (S1–S15: the two-floor backup, eleven-belief piecewise linearity, degeneration against brute force, 48-cell closed-form agreement, partition coincidence, witness-family loci, the exact deficit on all 42 nonviable cells, the class declaration by brute force, rationality of the 384 vectors, monotonicity, the oscillation identity to horizon 24, model completeness, the selector split, dimension padding, and the one-sided jump values). The hidden-parameter script runs six check families on the adaptive-observation instance (theta observed on all 16 cells, the probe threshold $\frac{21}{10}$ as a branchwise identity, the deadline identity at $T_{\text{learn}} = 0, \dots, 4$, injective revelation, the antitone kernel chain, and the unknown-regeneration two-patch instance). The master script for this paper chains the three, then recomputes independently from the definitions: the closed-form values on the whole 16×12 grid by direct branch-mass enumeration, the resulting census (36 unit values and 156 half values on the declared class), the identity of the class-difference set with the region $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$, checked on an extended grid to $z_0 = 4.0$ and equal to the timing-cell combinations on the certified grid at $k \geq 2$, by brute-force enumeration over blind sequences; the 384-vector type census (156, 156, 72) and its deduplicated 348-vector form; the hold-class deficit profile on all 48 cells (0 before the falling-branch exit, $\frac{1}{2}$ after); the lost-branch step of the asymmetric audit (step 1 for all four sequences, heavier branch under $u_1 = +1$, lighter under $u_1 = -1$); the pinned-probe learning-deadline chain $\frac{21}{10} + \frac{T_{\text{learn}}}{10}$ with the free-probe (21/10, flat) and probe-free ($1 + T_{\text{learn}}/10$,

kernels 16 down to 12) benchmarks; and every headline number asserted present in the text and bound to the scripts’ certified families. Two independent paths back the closed forms: a brute-force path that enumerates trajectories straight from the definitions (all holds and all open-loop sequences, pathwise survival) over a structured sweep of 2268 triples (z_0, w, ℓ) including $z_0 < 1$, and a seeded randomized battery of 500 further cases; the survivable-set antichain, the class ordering and its gaps, the stabilization freeze (spot enumeration to horizon 15 and value agreement to horizon 40), the weighted-path identity, the sensor counterexample, and the noisy-probe and majority closed forms are each re-derived by independent enumeration; the contamination lemma and the delayed-class ρ -closed form are checked against vertex enumeration, the probe-count bound at $n = 1, 3, 5, 7, 9$ in exact rationals, and the additive deadline law on both instances.

12. Delimitations and outlook

The semantics here is stochastic expectation: the value is a survival *probability*, and the robust quantifiers of the companion papers are recovered only at the deterministic degeneration, which is a change of operator (Remark 10); the stochastic and robust readings of a genuinely stochastic plant differ, and this paper does not conflate them. The timing layer’s statements, here and in the companion papers, are class-scoped (Remark 17). No learning theory is claimed beyond the one-probe exact split: Bayesian updating, repeated experiments, and exploration–exploitation trade-offs are out of scope; the deadline identity is for the pinned-probe schedule family (hold to the probe horizon, probe, learned act), with the free-probe and probe-free benchmarks stated alongside, and other schedules are not classified; the parameter θ enters additively, and state-dependent parameter dynamics belong to the hybrid companion’s setting. The alpha-set growth is exponential in the worst case — and finite-horizon partially observable problems are PSPACE-complete, their no-observation variant NP-complete (Papadimitriou and Tsitsiklis, 1987) — so the certified two-witness families are structural, not representative. The contamination interpolation of Section 10. is the formal home of worst-case wording; the moment- and distance-based ambiguity families (piecewise-linear values persist there; Nakao, Jiang, and Shen, 2021) are the natural next ring and are not developed here. Scaling past the certified instances — exact point-based value iteration with rational solvers, survivable-set compression over the antichain of Proposition 5, and medium instances with several hidden parameters — the probe-count asymptotics of the repetition trade-off, extensions of the learning-deadline law to misaligned worst-loss hidden parameters, are the natural next steps. A third item stood on that list when this paper was drafted — a proof-assistant formalization of selected combinatorial statements — and now covers the deterministic support identity under its declared kernel hypotheses, the freeze value for its specified action families, and rationality of the declared α -vectors. These are machine-checked (Lean 4; no extttsorryAx in the audited import environment, with generated extttnative_decide trust dependencies scoped below), the first on the repaired reading of Theorem 3 recorded in Remark 4. One bound remains cited rather than formalized: the Sperner bound of Proposition 5 (iii), for which the layer supplies the binomial machinery needed to *state* it but no proof. The deficit-bound reading of Proposition 13 remains the interface through which this theory attaches to the certificate families already established, and the exact rational values of the certified campaign are the benchmark any relaxation must meet.

13. Conclusion

The belief-state safety value carries the obstruction calculus into the probabilistic setting without loss of exactness. The support identity ties the value-one level to the viable-set recursion; rational α -vectors keep every certified value exact, and the complete value table shows the two-valued structure the certificates predict; the certificates survive as deficit lower bounds, attained on natural instances on both the symmetric and the asymmetric prior; the certified delayed-regime class agrees with its closed form cell by cell, with the class declaration responsible for exactly the 36 timing combinations on the grid; and the adaptive-observation audit prices late resolution exactly — for the pinned-probe family a linear learning deadline on the viable set, $|K| = 5, 4, 3, 2, 1$, bracketed by the flat free-probe threshold and the probe-free benchmark — while the stochastic layer prices observation noise exactly: a weighted-path deficit identity, a sensor instance breaking the min-mass bound’s universality, and a repetition trade-off at 7/250 per extra probe pair. The observation layer, not the forecast layer, sets the value; the design rules are the same ones the deterministic audits impose, now with a price attached and every number in the tables open to script-level re-inspection.

1 Related work

This paper asks when an observation-based policy is *sufficient*, exactly, and what remains safe when it is not. Both halves have near neighbours, and this section names them before stating what is claimed.

1. The belief-state reduction and the α -vector representation

That a partially observed problem can be written as a fully observed one on the space of beliefs is classical (Aström, 1965), and the finite-horizon value function of a partially observed Markov decision process is piecewise linear and convex in the belief, represented by a finite set of α -vectors (Smallwood and Sondik, 1973; see Bertsekas and Shreve, 1978, for the measure-theoretic setting). The safety value $V_k(b)$ studied here is a belief-space value in that lineage, with one structural change: the recursion maximises over actions but *minimises* over the disturbance, so the object is a worst-case safety probability rather than an expected total reward. Consequently the piecewise-linear representation survives but its identifying structure does not: here the α -vectors are exactly the indicators of the maximal jointly survivable subsets of the support, an antichain (Proposition 5), rather than the reward-derived vectors of the expected-criterion theory.

2. Sufficiency under imperfect measurement: the two nearest neighbours

The closest work is the viability-theoretic treatment of imperfect information, and it comes in two forms that this paper sits between.

Veliov (1993). For a time-invariant system with incomplete and inexact measurement, Veliov gives a **sufficient** condition for the existence of an output feedback regulation map, and shows it is equivalent to Haddad’s viability condition when the measurement is perfect. The condition is sufficient and not necessary: a system may admit an adequate output feedback that the condition does not detect.

Doyen (2000). (**Luc Doyen, the viability theorist — not Laurent Doyen of the qualitative partially-observed literature cited below; the surnames coincide and the two should not be conflated.**) For uncer-

tain nonlinear systems under state and control constraints with bounded disturbances on both the dynamics and the output, Doyen gives conditions that are **necessary and sufficient** for the guaranteed viability property — defined as the existence of a *Lipschitz closed-loop* maintaining the state *exactly* in a given closed domain — and derives equivalent geometric and Hamilton–Jacobi–Isaacs characterisations via the Lipschitz kernel of a set-valued map.

These two delimit the contribution. Relative to Veliov, what is added is exactness: the condition here is necessary as well as sufficient, so its failure is informative rather than merely inconclusive. Relative to Doyen, what differs is the admissible policy class and the criterion. Doyen’s loops are Lipschitz and *memoryless*, of the form $u(h(x, w))$; the recursion here admits policies that depend on the whole observation history, and the distinction is strict — the feedback viable-set recursion $\mathcal{W}_k^{\text{fb}}$ contains the blind recursion $\mathcal{W}_k$, strictly in general (Remark 4). And where Doyen’s criterion is the exact maintenance of a closed domain, the safety value here is quantitative: when a belief is not fully safe, the deficit satisfies $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$ (Proposition 13), so near-misses are measured rather than discarded.

3. The estimation-space and qualitative programmes

A distinct strategy absorbs imperfect information into an enlarged state: the estimation-tube programme formulates the problem on a space of estimators or tubes rather than on the belief simplex (Cardaliaguet, Quincampoix and Saint-Pierre, 2007, for the survey of viability methods in this line). That programme is set in continuous time and targets differential games; the present development is discrete, finite-horizon, and computational, and it does not compete with it.

On the qualitative side, the decidability and complexity landscape for partially observed Markov decision processes under qualitative objectives — reachability, safety, Büchi — is established (Chatterjee, Doyen and Henzinger, 2009), and it is known that almost-sure safety depends only on belief supports. What is added here is quantitative and structural: the deficit bound and its witness reading, the survivable-set (antichain) formula behind the witness census, the class lattice with its realized gaps, the parametric closed forms, and the class-scoped timing obstruction — together with the observation that the support-only reduction, valid for almost-sure objectives, does *not* extend to the quantitative criterion, which is why the value, not just the support, has to be carried.

4. Complexity, robustness and shielding

That finite-horizon partially observed problems are computationally hard is classical (Papadimitriou and Tsitsiklis, 1987), and the practical response has been bounds and approximations (Lovejoy, 1991). This paper does not improve those complexity results; it identifies classes on which exact answers are nonetheless available in closed form, and records where exactness stops. Distributionally robust formulations place an ambiguity set over the transition or the observation kernel (Nakao, Jiang and Shen, 2021); the disturbance model here is worst-case over a support rather than distributional, so the two are not comparable and neither subsumes the other. Safe reinforcement learning via shielding (Alshiekh, Bloem, Ehlers, Könighofer, Niekum and Topcu, 2018) enforces safety by constraining a learner’s actions; the setting there is learning with an unknown model, here the model is known and the question is exact sufficiency.

5. What is claimed

Given the above, the contribution is five things and nothing more.

1. **Exact sufficiency where the nearest sufficient condition is only sufficient.** The value-one level of the recursion coincides with the feedback viable-set recursion, so the condition is necessary as well as sufficient (Theorem 3).
2. **Admission of history-dependent policies**, with the feedback/blind distinction shown to be strict (Remark 4) — a repaired reading; the mechanized support identity is checked on the repaired statement.
3. **A quantitative deficit rather than a binary verdict:** $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$, with the witness reading of Proposition 13.
4. **The structural identification of the α -vectors** as the indicators of the maximal jointly survivable subsets of the support, Sperner-bounded (Proposition 5), the class lattice with realized gaps (Theorem 15), and parametric closed forms on named classes.
5. **Machine-checking:** the support identity, the freeze count and its frozen value, and the rationality of the α -vectors are formalized in Lean 4 with no extttsorryAx in the audited aggregate-import environment. Three extttnative_decide uses in the P3 support/value examples generate nine axiom names in this pinned source, relying on compiled finite computation rather than kernel-only reduction. One bound remains cited rather than formalized: the Sperner bound of Proposition 5(iii).

Current Lean provenance. The committed project pin v4.34.1 was rebuilt in GitHub Actions (default Lake target: 60/60 jobs, exit 0; <https://github.com/MIKEAA2020/general-sustainability/actions/runs/36946730855>). The exact project source and two footprint-check .lean files are archived with SHA-256 manifests. A comment-aware scan of all 62 source files finds no sorry, admit, explicit axiom or constant declaration. Twenty-one named footprints contain no sorryAx; the aggregate-import environment exposes three source-level native_decide uses in the P3 support/value examples, yielding nine distinct generated axiom names under this pin. These are executable computational certificates that extend kernel-only trust to the Lean compiler/runtime; they are not admitted gaps and must not be described as zero axioms or as kernel-only proofs. Three preserved historical modules outside the aggregate import closure are not certified by the default build. The pinned declared-exceptions gate passed in GitHub Actions (run 36990482599): only the nine exact generated names from three approved P3 source sites are permitted; any change to those sites or names requires separate review and re-approval. The separate strict zero-exception check exits 2 by design, while the build itself passed 60/60; the two verdicts must not be conflated. The v4.14.0 comparison of the current source fails in the prelude and does not reproduce the old source run.

Not claimed: the belief-state reduction (Åström), the α -vector representation as such (Smallwood and Sondik), the qualitative complexity landscape (Chatterjee, Doyen and Henzinger), or the hardness results that motivate approximation (Papadimitriou and Tsitsiklis; Lovejoy).

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Declarations

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None.

Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_probabilistic_sufficiency_v14_verification.py` — which chains the per-system scripts `paper2_belief_state_v2_verification.py`, `paper2_stochastic_selector_v2_verify.py`, and `hidden_parameter_learning_v1_verify.py` — and the figure-and-table generation script `paper2_belief_state_figures.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`, at the commit pinned to this article’s submission); they reproduce all values, bounds, thresholds, figures, and tabulated entries verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.